

Q2 2024 Earnings Call

Company Participants

- Manish Bhatia, Exec VP:Global Operations
- Mark J Murphy, Exec VP/CFO
- Satya Kumar, VP:Investor Relations & Treasury
- Sumit Sadana, Exec VP/Chief Business Officer

Other Participants

- Aaron Rakers, Analyst, Wells Fargo
- Brian Chin, Analyst, Stifel
- Chris Caso, Analyst, Wolfe Research
- Eddy Orabi, Analyst, TD Cowen
- Karl Ackerman, Analyst, BNP Paribas Exane
- Mehdi Hosseini, Analyst, Susquehanna
- Nicholas Doyle, Analyst, Needham & Company
- Thomas O'Malley, Analyst, Barclays
- Timothy Arcuri, Analyst, UBS

Presentation

Operator

Thank you for standing by. Welcome to Micron's Post Earnings Analyst Call. At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions). As a reminder, today's program is being recorded. And now I'd like to introduce your host for today's program, Satya Kumar, Corporate Vice President, Investor Relations and Treasurer. Please go ahead, sir.

Satya Kumar {[BIO](#) [18241786](#) <GO>}

Thank you, and welcome to Micron Technology's Fiscal Second Quarter 2024 Post Earnings Analyst Call. On the call with me today are Sumit Sadana, Micron's Chief Business Officer; Manish Bhatia, EVP of Global Operations; and Mark Murphy, our CFO.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, market trends and drivers, our expected results and guidance, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today. We refer you to documents we filed with the SEC, including our most recent Form 10-Q and upcoming 10-Q for a discussion of risks that may affect our future results.

Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, and achievements. We are under no duty to update any of the forward-looking statements to conform these statements to actual results. We can now open up the call for Q&A.

Questions And Answers

Operator

Certainly. One moment for our first question. And our first question comes from the line of Thomas O'Malley from Barclays. Your question, please. Thomas, you might have your phone on mute. All right. I think we will move on. And Thomas, you can get back into the queue as time allows. One moment for our next question. And our next question comes from the line of Krish Sankar from TD Cowen. Your question, please.

Q - Eddy Orabi {BIO 22349154 <GO>}

Hey, guys. This is Eddy for Krish. Thanks for setting up this call back. Two-part question for Sumit, if I may. The biggest bottleneck for HBM supply today is probably in back-end capacity. Please correct me if I'm wrong. In case your market share opportunity in HBM turned out to be better than you guys are expecting, do you see a scenario where back-end capacity may limit the opportunity next year?

And two, obviously, you have HBM taking away most of the CapEx dollars today and you have higher DRAM gross margins demand. So it seems DRAM and HBM may continue to consume the majority of the CapEx in the next few quarters. But do you see a scenario where you increased net CapEx this calendar year based on pricing action in that market? Or are you at a point where you think it's still early to invest in NAND despite the improving profitability prospects of that market? Thank you.

A - Sumit Sadana {BIO 7009258 <GO>}

Yes. So I'll make a couple of comments, and then I'll also request Manish to jump in and talk about some of the items you mentioned. So first, at a company level, our goal is to maintain our bit share in both DRAM and NAND at fairly consistent levels. So we target flat consistent overall bit share in both DRAM and NAND. And it's not just one or the other but both. So we'll continue to invest in a very disciplined fashion with a close eye on how the demand is shaping up to ensure that our supply and our CapEx investments enable us ultimately to have that flat bit share.

HBM is a very exciting product. We have an industry-leading product here, best performance specs, 30% lower power, a lot of customer demand, more customer demand than we know how to meet. And you have heard already in our prepared remarks and in the following Q&A that 2024 is sold out, 2025, overwhelming majority already allocated.

So we are going to obviously continue to be very focused on disciplined investments and CapEx, whether it's for HBM, whether it's for DRAM, whether it's for NAND to ensure that our target of sometime in '25 getting HBM share to be equal to DRAM share, we get to that point. And as regards to your comments about where some of the bottlenecks in HBM are? How we look at those things, I'll request Manish to comment on some of that.

A - Manish Bhatia {BIO 18045807 <GO>}

Sure. So I think we're -- I want to start -- I mean HBM, it's -- our HBM3E is an industry-leading product, Eddy. And one of the backbones of that is the fact that it's built on our 1-beta process. And we've had -- I think, on this call, 12 months ago, we probably talked to you about the fact that we had reached our sort of mature yield milestone on 1-beta so a year ago. So it's the fact that we're ramping HBM3E on a very mature, strong industry-leading platform, gives us a lot of confidence in our ability to build the HBM wafers.

You're right that we are ramping our assembly and test capacity to be able to -- now that we've productized the -- we've begun production shipments. We announced today that we started production shipments for our customer and the ramp is on for us throughout this year and into

next year. We've made really good progress on our yields and on our capacity and the throughputs within that capacity, we feel very good about. And feel good about being able to reach the targets that Sumit just talked about with regard to having our share of HBM match our overall -- our share of overall DRAM capacity in bits sometime next year.

In terms of your question on CapEx just to Sumit's point, we make these decisions with regard to how much CapEx we'll spend in DRAM or NAND, both based on the demand that we see, but also independently, right? So I think we'll be looking at the market for -- and the demand for DRAM across all the drivers. And then similarly looking at the market for NAND and trying to decide how much we would invest there to be able to meet the demand that we see in that market separately.

Q - Eddy Orabi {[BIO 22349154](#) <GO>}

Got it. Thank you, guys.

Operator

Thank you. One moment for our next question. And our next question is from Thomas O'Malley from Barclays. Your question, please.

Q - Thomas O'Malley

Hi, guys. Thanks for looping me back in. I appreciate it. Having a little bit of connectivity issues here. But mine is a two-part. One is I think capacity constraints was a common theme on the call that kept coming back up again. Could you talk about what is causing the capacity concern? Is it just the time it takes to put tools in your fabs in order to build HBM? Is it potentially a tool that you're unable to get? Just any kind of color on what is the limiting factor there? And if there could be some potential upside if something were to go a bit better?

And then the second part of the question is you obviously have a public announcement with NVIDIA. But just given your capacity constraints, how should we be thinking about your ability to serve some of the other AI accelerator names in the market next year as well? Thank you very much.

A - Manish Bhatia {[BIO 18045807](#) <GO>}

So thanks, Tom, and welcome back. So I'll take the CapEx part, and then I'll let that -- and capacity part, I'll let Sumit take the customer-related question. So really, there are 3 factors, I think, in terms of the -- what's creating the supply tightness at leading-edge nodes. And 2 of them are common to DRAM and NAND, and then the last one is really unique to DRAM. But the first one is that we, as well as the rest of the industry have had significant capital reductions as we went through the downturn over the last 18 months.

And so those -- the lower basis of CapEx that we've had and we've -- I mentioned on the previous call WFE for us is down fiscal '24 over fiscal '23. So those decisions and the timing of that certainly impact our supply on the leading edge nodes, and it was done, obviously, as we have had very, very high inventory levels. So we're trying to make sure we time any increases in CapEx as in line with inventories coming down.

And the second is this comment that we've been making, and we've been trying to be transparent that we see -- we have undertaken a strategy to structurally reduce the wafer start capacity across both DRAM and NAND. And we're doing that to be able to take some of that

capacity and equipment and actually apply it to converting to these leading-edge nodes. It's a capital-efficient way to get the benefits of these new technologies.

These new technologies in and of themselves have lower cost, but they also have better performance, lower power, in particular, 1-beta. Again, that's the building block for our HBM has higher performance and lower power than others in the industry and certainly than our prior node. But that structural reduction also reduces our capability overall to -- on supply. And those 2 factors are common to DRAM and NAND.

The third one, of course, is the trade ratio for HBM as we increase the mix of HBM in our business. And as HBM increases in penetration across the industry, this trade ratio of needing to start for HBM3E, approximately 3 wafers in order to get the same number of bits as you would, were you to be starting at one DDR5 wafer. This also weighs on the supply capacity for everyone in the industry, frankly.

So that's sort of a little bit of a breakdown on sort of what's creating this. And of course, our levels of profitability are still below where they need to be, even with the results we reported. So we are trying to be disciplined and make sure that we're able to return good ROI on any increases in capital investment that there would be, say, in fiscal year '25. Sumit, do you want to talk about that?

A - Sumit Sadana {[BIO 7009258](#) <GO>}

Yes. Sure. So I'll talk about the customer angle. We have been engaged with practically all of the key purchasers of HBM in the world. And as you know, there aren't that many of them. There are just a handful who account for the substantial majority of the TAM for HBM. And our engagements relate to all of the product road map, certainly, HBM3E, but also closely engaged on HBM4 and the full future developments that relate to that. And of course, on HBM3E, it is 8 high, 12 high, which, as we have mentioned, we have already shipped samples of 12 high as well for HBM3E. Now we have made the announcement with NVIDIA, of course, but you know that we are in the early stages of a substantial capacity ramp in HBM3E. And as we get to more capacity, you will (Technical Difficulty) to new customers, our goal is very clear that in 2025, we'll have a more diversified customer base for our HBM products. And we are making great progress towards that with the allocation to customers for the substantial majority of our 2025 capacity. So we are in really good shape and continuing discussions with the rest of the customers.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Mehdi Hosseini from Susquehanna International Group. Your question, please.

Q - Mehdi Hosseini {[BIO 4362002](#) <GO>}

Yes, Sir. Thanks for taking my question. You guys have been talking about 2025 as becoming a record year. And the team and Sanjay have consistently been talking about this for quite some time. What I wanted to follow up here is, what happens to the overall margin profile? We understand that you may not be able to share with us a trajectory of a price increase. But when you think about what you're doing with the cost structure and the product mix, does that give confidence that perhaps on a gross margin, you can at least meet the prior peaks? Or -- and if you cannot confirm it, can you talk about the dynamics of the margin profile in a confined of a record year in 2025?

A - Mark J Murphy {[BIO 16136048](#) <GO>}

Maybe I'll start, Mehdi, and Sumit, and Manish can continue to. You may want to go on mute, Mehdi. (inaudible) There we go. I tried to lay out on the general call or the main call the arc of gross margin over the next few quarters, and then how that extends into next year. Price has been the dominant factor in the margin expansion over the last couple of quarters and including the second quarter of 19 points. And then this increase in second quarter to third quarter of 650 basis points. Price is the dominant factor. But increasingly, price, while it will continue to increase into '25 and we believe through '25, other factors will contribute to margin expansion.

Favorable mix effects through a number of value-added products that you heard about today, first and foremost, HBM. And then also just the resumption of more normal cost downs, excluding this HBM effect because the period costs are coming off and the node transitions are occurring and so forth that drive those costs down. So we can't -- we'll provide more specific gross margin guidance on '25 on a future call. But those factors of increasing price, improving mix, cost competitiveness, those factors will contribute to margin expansion '24 to '25.

I will comment a bit more on fourth quarter. We do expect bits to be up slightly for DRAM, 3rd to 4 somewhat for NAND. And then for the remainder, of course, this calendar year, we would expect sort of flattish bit shipments, but somewhat higher mix of NAND. So I just think about that as you model gross margins the next quarters beyond third quarter. We do see, again, the improvement of stronger mix of leading-edge products and richer mix. And so we would expect in the fourth quarter to be somewhat above 30% on gross margin, and that includes, obviously, the effects of higher pricing. But keep in mind that higher NAND mix.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Chris Caso from Wolfe Research. Your question, please.

Q - Chris Caso {BIO 4815032 <GO>}

Yes, thanks. Good afternoon. The question is about some of the commitments that you spoke about on the call with respect to HBM. Could you go into a little more detail on that? It wasn't clear on the call whether those pricing and volume commitments extended into calendar '25 as well. And obviously, as Sanjay said in the call, it's not something that's very usual. What's driving that? And does it have any effect in capacity planning in that -- at least the HBM part of the business, you're getting better visibility as compared to normal?

A - Sumit Sadana {BIO 7009258 <GO>}

Yes. I mean we do have better visibility into HBM for a couple of reasons. Number one, the growth in the industry is pretty substantial. So the HBM TAM and overall opportunity is growing rapidly. And you've seen even earlier this week, the Blackwell GPU has 33% more HBM attached and every new GPU that has been announced by NVIDIA had starting from A100, 40 gigabytes of HBM to H180 gigabytes, H200, 144 gigabytes, and now Blackwell at 192 gigabytes. So there has been a substantial increase in HBM and hence, the TAM is going up and gives us good visibility there. The other perhaps even more important part is we are really feeling good about our HBM3E product. We spoke about the leadership capabilities of this product. Not only does it have really high performance, but all of the testing that has been done with customers shows it has tremendous headroom on the performance side for -- even going beyond the rated specs. And we have 30% benefit on power consumption and all of you know how much of an impact that makes on data centers from both an operating cost perspective as well as just having the constraints on power define the specs of the data center sometimes. So these are really good capabilities. And because of that, we are seeing a lot of preferential demand from customers for

our HBM3E product. So in terms of the unusual nature that Sanjay mentioned, being CQ1 for the overwhelming majority of our 2025 capacity that we are intending to put in place to be allocated is a very unusual thing in our business. And we are not only able to have these agreements done with customers, but some of these have already included price being negotiated for 2025. Of course, 2024, volume and price is all done for the full year. In 2025, some of the agreement has priced already done. The rest of it will be done. My expectation is fairly soon. So the other aspect I will mention is that these agreements tend to have more teeth in them, more stringent terms in terms of commitments from customers, and so that's another positive. And I think that's all consistent with longer lead times, more complex products and also the very good capability that the Micron team has created with this product and all of this is a testimony to that.

A - Manish Bhatia {[BIO 18045807](#) <GO>}

And just to your point, Chris, yes, it means having this much visibility certainly does help with supply planning and justifying the capital that we need to put in place. And we talked about the margin structure for HBM3E being accretive already to DRAM. So -- but the overall DRAM -- our overall company profitability is still not where we'd like it to be. So we're being very disciplined and careful and investments we'll make for the rest of the business.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Karl Ackerman from BNP Paribas. Your question, please.

Q - Karl Ackerman {[BIO 19693285](#) <GO>}

Yeah. Thank you, gentlemen. I was hoping you could discuss which areas of the market you're seeing the strongest rebound in volume both in the May quarter as well as the August quarter? And secondarily, do you see a stronger-than-expected recovery in pricing in DRAM, excluding HBM and NAND over the next couple of quarters? Thank you very much.

A - Sumit Sadana {[BIO 7009258](#) <GO>}

Yeah. So in terms of volume, I think, if I may just zoom out a little bit, over the last couple of quarters, we had mentioned to you that our expectation was that PC OEMs and smartphone OEMs would be purchasing first, because they were the first to go into a downturn and that is exactly what had happened. That's where the strength had been focused in the last few quarters.

And we had also mentioned to you that our expectation was that sometime in the first half of calendar 2024, data center will start getting normalized in terms of inventory and more normal buying patterns would slowly return in the data center. And right on cue, we have seen in the last many weeks, last couple of months, upsides in data center from our customers. And these upsides have serve to tighten the market.

Of course, the PC volumes and the smartphone volumes expectations for calendar 2024 haven't really changed much in the last several months. They are pretty much very similar levels as what it was for end unit and sell-through of units. But data center is stronger. The traditional data center server sales are starting to improve. We're getting upside on the DRAM side as well as on the data center SSD side.

And we're also starting -- we are also seeing obviously exceptionally strong demand coming from AI servers. And these AI server demand, of course, there is the HBM angle, which we have discussed at nauseam, but very strong DDR5 demand, and our DDR5 has certainly far more

demand than we have supply. So we are chasing supply there on very tight on the leading edge there. And of course, the strength in the data center SSD is also very notable.

So a couple of data points that we shared in our prepared remarks that I just wanted to highlight to you. For example, robust growth in DRAM sequentially in the data center in FQ2, cloud revenue in FQ2, doubling -- more than doubling sequentially. And data center SSD has been performing really well for us year-on-year, up over 100% in FQ2. And we have been hitting new record share levels as our share of data center SSD continues to hit new peaks for the last several quarters.

So based on all of this, I think the pricing has been in a good trajectory for both DRAM and NAND. And the strength in the data center has served to tighten the supply situation, supply-demand balance, which is pushing up pricing and having that ripple effect we mentioned across all segments. And so that's how we look at it.

And of course, some of the focus that you saw from us on CapEx, we haven't changed our view on CapEx. We remain very disciplined in our CapEx. We remain very disciplined in our supply planning. Our goal is to substantially improve our financial performance before we commit any more capital, and that's the approach we are taking.

And you know that the lead times are very long in this business. You order the tool to expand capacity 9 months plus to get the tools and then you get cycle time through the fab and through the back end and another 5 months, we're looking at 14 months from the time you order tools to the time you get output. So that's really part of what was behind our expectation that we expect the pricing to continue to strengthen through calendar '24.

Q - Karl Ackerman {[BIO 19693285 <GO>](#)}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Aaron Rakers from Wells Fargo. Your question, please.

Q - Aaron Rakers {[BIO 6649630 <GO>](#)}

Yeah. Thanks for doing the follow-up call. I guess my one question and I'll leave it at that, is just as you see the fundamental improvement in the business and clearly a path to sustainable or appears to be sustainable free cash flow. Mark, I'm just curious how you're thinking about capital return. And it's been a while since we've got to talk about share repurchase. But just curious of how you think about share repurchase activity as the model moves forward, and looking out over the next couple of quarters. Thank you.

A - Mark J Murphy {[BIO 16136048 <GO>](#)}

Yeah, it's a good question to think about, Aaron. As you say, we're pleased with the improved profitability and outlook and also the continued working capital performance CapEx discipline that Sumit was talking about, so pleased to be near breakeven free cash flow. And then as you heard on the call today, expect free cash flow generation in third quarter and fourth quarter.

I think the balance sheet remains the highest priority. And fortunately, it's in great shape. We've still got very high liquidity, which we maintained through the downturn. I'll note our leverage has peaked prior quarter and has been coming down -- started coming down this last quarter. And it's

always been low on a net basis. But we're -- again, we're in a great spot there and, of course, have absolute commitment to investment grade.

Over time, as the free cash flow generation strengthens and as we said in our Investor Day, we intend over time to grow the dividend and then return excess cash to shareholders. So we're very much consistent with what we said at Investor Day and we'll be in a position at some point to make those decisions.

Q - Aaron Rakers {[BIO 6649630 <GO>](#)}

And Mark, is there a liquidity level that you kind of think about, or excess cash level you kind of think about?

A - Mark J Murphy {[BIO 16136048 <GO>](#)}

Yeah. We've put out there at Investor Day, we had this 35% of sales, but I'd say we have more than ample liquidity now. And again, it improved steadily from -- we maintained high liquidity through the downturn. Obviously, the outlook has been improving. And we're in the mode of generating free cash and we'll just take in a number of factors as we go forward. But nothing's changed from our commitments we laid out at Investor Day.

Q - Aaron Rakers {[BIO 6649630 <GO>](#)}

Fair enough. Thanks, Mark.

Operator

Thank you. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Thanks. Mark, inventory was up. I kind of expected it to come down. I think you also thought that it would probably come down a little bit. And I know you said finished goods are down. So I guess, is that WIP is up? And I assume it's mostly related to the HBM ramp. Can you kind of talk about that?

A - Mark J Murphy {[BIO 16136048 <GO>](#)}

Yes, you've hit on some of the things already. I mean in the end, it's -- listen, it was around where we thought it would be. It's -- if you adjust for the low-cost inventory that was in there, we were actually down on dollars and DIO. So I think inventories continue to track is consistent with what we thought. And importantly, we still see them going down on a DIO basis through '24 and into '25. We think we'll be in a real good position relative to our target levels by end of '25. So again, completely in control and things happening there as we expect.

We're extremely tight on leading edge. In fact, a decent amount of what sold out in finished goods is DDR4. And then as you talk about what's building in WIP, is a sizable part of that is HBM and some other leading-edge products. So again, in control kind of what's happening in inventory is what we would like to be happening. At a macro level, it's going down. At a more tactical level, it's mixing as we need for the business and feel good about it.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

Yes. And I'll just add, it's not just HBM, but as we transition the portfolio to higher-value products, Tim, right? So data center SSD is another good example. And these higher-value products are system-level products, they do have longer cycle time. And so there will be a sort of the lengthening of the supply chain for that just as we gain more traction. Sumit talked about how we're doing it -- how well we're doing in data center SSDs and how broad our future looks there.

HBM also longer cycle time and assembly and test as well. So that will be something. But at the same time, we're definitely focused on across the board, trying to make sure that as we kind of have this longer-term visibility from our customers, with them committing to demand with longer lead times. And as part of these LTAs, obviously, that gives us the ability to be able to be more lean as we do run things through the supply chain. So that's why Mark and I are confident that we're going to gradually be working down inventory as we go through the next few quarters.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Got it. Okay. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Nicholas Doyle from Needham. Your question, please.

Q - Nicholas Doyle {[BIO 1922054](#) <GO>}

Hi, guys, on for Quinn Bolton. Thanks for letting me ask the question. On the call, you talked about total industry server unit shipments growing at mid-to-high single digits in calendar '24, I think that's in line with consensus. Can you expand on the quarterly trajectory? Given your customer conversations, do you see a significant ramp in the second half? Or is it more stable through the year? Maybe traditional CPU coming back drives a second-half acceleration? And any thoughts on how you're thinking about '25 server units would be helpful. Thanks.

A - Sumit Sadana {[BIO 7009258](#) <GO>}

Yes. In terms of the server units, the first quarter was not great. First calendar quarter, 2024 was not great for traditional servers but we started to see some of that demand starting to come back late in the first quarter. So we do think that Q2, it improves from Q1 sequentially, and then second half of the calendar year is stronger than the first half.

And it helps that the economy seems to have better footing as well. Most of the concerns around recession, et cetera, that were there last year have abated. And so the traditional server demand is starting to improve. AI servers are exceptionally strong. Our view is it is more supply-limited in terms of how much AI servers can ship, but the demand for traditional servers should continue to improve through the calendar year. With 2025, we have an optimistic view that 2025 will be the first year in several years when the data center will be buying robustly for the full year because if you think about 2024, it's still more of the first half is when the inventory normalizes sometime in the first half of calendar '24.

So the buying patterns would be starting to get normal only in the second half of 2024. So 2025 is the first full year in a while for data center to come back, both the traditional data center purchases as well as continuing robust ramp of AI servers in 2025. So we expect '25 to be a very strong year for the data center overall across both DRAM and NAND and across both AI servers and traditional servers.

Q - Nicholas Doyle {BIO 1922054 <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Brian Chin from Stifel. Your question, please.

Q - Brian Chin {BIO 5763426 <GO>}

Hi there. Thanks for taking the question. Maybe I'll kind of ask a little bit about cost road map here. Regarding the DRAM wafer conversion loss, can you touch on how we went from, I think, a 2:1 ratio to an even more severe 3:1 and also, if the die size is 3x, is cost also 3x, or what's a better way to look at that? And then just a second part to that, what happens to the cost down when you start ramping 1-gamma? And is it likely that we won't see our 1-gamma qualified for HBM until HBM4?

A - Manish Bhatia {BIO 18045807 <GO>}

Okay. So Brian, a couple of questions there. Let me start with the last one first, which is that we're not really commenting on HBM4 yet. We feel good about 1-gamma overall. 1-gamma is on track, and we had some comments in the script around it coming to market for us in calendar year '25 and starting to ramp supply of that in calendar year '25, and it's making good progress.

In terms of the -- your question about how we went -- how the trade ratio for HBM3E how we arrived at 3. I think we've always said it was more than 2. And I think we're just kind of providing more color on where we're seeing it landing now. It's a -- there are a few different factors there in that trade ratio. One is the die size between the HBM die and a standard DDR5 die. The second is the yield trade-offs that are there in trying to be able to deliver a high-performance product with such a high die stack and a complex in particular, complex assembly and test process.

And then the third is the fact that we now have a logic interface wafer that we need to produce as well, which also takes DRAM capacity. So we're including all of those in our trade ratio, and we do believe that this is similar for us with everyone else in the industry because the HBM die is standard across the industry based on the GenX standard. And keep in mind that even with what we've talked about today, for us, as we're starting to ship production revenue this quarter, it's already margin accretive versus the rest of DRAM. So we are getting a pricing premium for the higher costs that we're incurring to build the HBM product.

Q - Brian Chin {BIO 5763426 <GO>}

Sure.

A - Mark J Murphy {BIO 16136048 <GO>}

Maybe just -- maybe, Brian, on that question, maybe just some housekeeping on cost downs. We do -- as we mentioned on the call, that the FY '24 front-end cost ex HBM, as we've said many times, is mid-to-high single digits for DRAM and then NAND low double digits cost downs for NAND. So FY '24, all-in for DRAM, we would expect it to be similar. But what we're seeing is the benefit from this period costs and all these inefficiencies coming off is helping offset the HBM mix initially. Now eventually, that HBM mix is sequentially going to become -- if you're not going to have that favorability associated with the period costs rolling off, you're going to have an

increasing volume of HBM. So you're going to see that, again, dampen cost downs and contribute to cost. But as Manish said, good trade given the favorability of that product.

Q - Brian Chin {BIO 5763426 <GO>}

And Mark, when you said the all into the -- I mean front end and back end all in for the year be a similar range to what we said the front end mid to high single digit ranges.

A - Mark J Murphy {BIO 16136048 <GO>}

Correct.

Q - Brian Chin {BIO 5763426 <GO>}

Obviously. Including HBM cost would be somewhat...

A - Mark J Murphy {BIO 16136048 <GO>}

And then on NAND, I mean a little lower on NAND all in as Manish said, considering all the components and back end would be high single digits, cost downs for FY '24, because we've got larger component SSD NAND. So again, premium products and good trade on that incremental cost.

I think as we look out to '25, we're obviously forecasting HBM to ramp in fiscal '25, so that would impact cost downs in DRAM. And then likewise in NAND, you're still going to see is -- this SSD mix and other things would, again, -- would be dampening cost down effects, but again, it's a value-added fixed rate.

Q - Brian Chin {BIO 5763426 <GO>}

All right. Great. Thanks, Mark. Thanks, Manish.

Operator

Thank you. This does conclude the question-and-answer session, as well as today's program. Thank you, ladies and gentlemen for your participation. You may now disconnect. Good day.

This transcript may not be 100 percent accurate and may contain misspellings and other inaccuracies. This transcript is provided "as is", without express or implied warranties of any kind. Bloomberg retains all rights to this transcript and provides it solely for your personal, non-commercial use. Bloomberg, its suppliers and third-party agents shall have no liability for errors in this transcript or for lost profits, losses, or direct, indirect, incidental, consequential, special or punitive damages in connection with the furnishing, performance or use of such transcript. Neither the information nor any opinion expressed in this transcript constitutes a solicitation of the purchase or sale of securities or commodities. Any opinion expressed in the transcript does not necessarily reflect the views of Bloomberg LP. © COPYRIGHT 2026, BLOOMBERG LP. All rights reserved. Any reproduction, redistribution or retransmission is expressly prohibited.